

Solution Requirements

Date	30 September 2026
Team ID	6ab39579211fa4fd5d28901b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>No login is required in the current EduGenie MVP. Students access the application directly through the web interface.</i>	Low
2	Authorization levels	<i>No role-based authorization is required because the current MVP does not provide authenticated multi-user roles.</i>	Low
3	External interfaces	<i>Google Gemini API for Q&A, quiz generation, summarization and personalized learning paths. LaMini-Flan-T5-783M is used locally for concept explanation.</i>	High

		<i>No external database is used. Gemini API is the external service used by this application.</i>	
4	Transactions processing	<i>Receives student input, validates the request, routes it to the appropriate feature/model, validates the result and returns the response to the interface.</i>	High
5	Reporting	<i>Provides generated answers, explanations, quizzes, summaries and personalized learning paths. Persistent report export is not included in the MVP.</i>	Medium
6	Business rules, etc.	<i>Routes each feature to its assigned model and validates required inputs and structured AI outputs before returning results.</i>	High
7	Compliance to laws or regulations	<i>Keeps the Gemini API key in environment configuration and excludes it from Git tracking. The MVP does not claim certification under a specific regulation.</i>	High
8	<i>Other</i>	<i>Provides five core learning features through a single responsive web interface with a FastAPI backend.</i>	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>Requests should return generated results without unnecessary interface delay under normal local execution and network conditions.</i>	<i>Feature requests return a visible result without frontend errors.</i>
2	Scalability	<i>The modular FastAPI structure allows additional learning features and model integrations without redesigning the application.</i>	<i>New feature modules can be added while existing routes remain functional.</i>
3	Security & Data Privacy	<i>The Gemini API key is stored in .env and excluded from Git. User inputs are validated before AI calls.</i>	<i>API key is not exposed in source control; invalid required inputs are rejected.</i>
4	Reliability & Availability	<i>Backend validation and service errors are handled with controlled responses so an individual request failure does not crash the application.</i>	<i>A request failure returns an error response and the application remains usable.</i>
5	Usability & Accessibility	<i>The interface clearly separates Q&A, concept explanation, quiz, summary and learning path features.</i>	<i>All five features are reachable and generated results are readable.</i>
6	Other	<i>Feature logic remains modular and model assignments remain centralized for easier maintenance and future extension.</i>	<i>Feature code remains separated into modules and model assignments remain centrally defined</i>